
Task W02: Backtracking — Answer Sheet

Name: _____

Before You Start

This is a LaTeX source file, not a finished document; what you are reading right now is what it looks like once compiled. To edit it and produce your own PDF, you will need a LaTeX environment such as Overleaf, TeXworks, or a local TeX installation; if you need help setting it up in Linux, please let me know.

1. Keep `Task_W02_Answers.tex` and `coa-accessible-student.sty` in the same folder. This file will not compile without the style file present.
 2. Open `Task_W02_Answers.tex` in a text or code editor, not a word processor, and replace each *italicized* placeholder with your own answer, following the instructions given in each section.
 3. Compile it by running `pdflatex Task_W02_Answers.tex`. Run this command twice in a row; the page numbers at the bottom of each page will not resolve correctly on a single pass, and will show "Page 1 of ??" instead of the real total.
 4. Every time you make a change to the `.tex` file, save it and run `pdflatex` twice again to see your latest edits in the PDF you submit.
 5. Submit both files: your edited `Task_W02_Answers.tex` source and the `Task_W02_Answers.pdf` it produces.
-

1 Problem 1: A Smaller Board

1.1 Board

Fill in the board below with a Q in each square where a queen ends up. Columns are 1 through 4 left to right; rows are 1 through 4 top to bottom.

The table has one line of source for each row of the board. Find the line that starts with the row number you want, then count across its blank cells the same way you count across the column headers above the table, and type a capital Q into the correct one. Leave every other cell in that line blank. Do not add, remove, or reorder any `&` symbols or the `\\` at the end of a line, since that changes the shape of the table.

For example, to place a queen in row 2, column 3, change this line:

<code>2 & & & \\</code>

to this:

<code>2 & & Q & \\</code>

	1	2	3	4
1				
2				
3				
4				

1.2 Backtrack Notes

One line for each column where your first-tried row did not work.

- Column ___, tried row ___ first, failed because ...

2 Problem 2: Row by Row

2.1 Your Rewritten Pseudocode

If you used the AI-for-formatting exception described in the assignment, submit the AI conversation as its own file alongside this answer sheet.

% Write your rewritten pseudocode here.

2.2 Board

Fill in this table the same way you filled in the one for Problem 1, using a capital Q in the correct cell of the correct row's line. This time the queens come from your rewritten row-by-row version, starting from row 0.

	1	2	3	4
1				
2				
3				
4				

2.3 Backtrack Notes

One line for each row where your first-tried column did not work.

- Row ___, tried column ___ first, failed because ...

3 Problem 3: Watch and Answer

3.1 Question 1

In your own words, explain what backtracking means for a knight's tour search, using the words "dead end" and "undo."

3.2 Question 2

Would a corner start need more backtracking than a central start, less, or is it impossible to say without trying? Explain your reasoning.

3.3 Question 3

Could changing the fixed move order matter here, the way it did in Problem 2? Explain your reasoning.

3.4 Question 4

Could a backtracking search ever find a complete tour on a three by three board? Explain your reasoning.

4 Reflection Questions

4.1 Question 1

Compare your Problem 1 board to your Problem 2 board, along with your backtrack notes for each.

4.2 Question 2

Compare your Problem 1 backtrack notes to the Knight's Tour worked example's 31 calls and 19 backtracks.

4.3 Question 3

Look back at your answer to Problem 3, question 2. Does that reasoning match what you saw across your Eight-Queens work?

4.4 Question 4

Would changing the row order within each column for Problem 1 behave more like Problem 2, or more like the Knight's Tour example? Explain your reasoning.

5 Bugs or Issues You Found

If you found something in the assignment that needs to be debugged, for example an unclear problem statement or an example that does not behave the way it is described, describe it here. Leave this blank if nothing came up.